

Chapter four

4. Computer Networks and Communications

4.1. Introduction to computer networking and its applications

Communication is a process of sharing ideas, information and messages with others in particular time and space. It includes writing, talking and non verbal communication like facial expressions, visual communications, electronic communications, etc.

Communication is also the transfer of information from one place to another. It may be manual one person talks to another. Or it may be mechanical or electronic gadget or device. Whatever the case, the process of communication involves the following **basic components**:

Source- Generates data to be transmitted

Message -Information/data to be transmitted

Transmitter- Converts data into transmittable signals

Transmission System- A Communication Equipment (intermediate device) that carries data

Receiver-Converts received signal into data

Destination-Takes incoming data

Protocol-Rules and standards that govern data communication

4.1.1. What is a Computer Network?

A **Computer Network** is a group of two or more interconnected computer systems that use common connection protocols for sharing various resources and files. You can establish a computer network connection using either cable or wireless media. Every network involves hardware and software that connects computers and tools.

Networks allow users to share peripheral devices (peripheral devices are nothing but input – output devices connected to the computer), program and data.

4.2. Types of Computer networks

Based on the size and the geographical areas, the Networks are classified into three types:

1. Local Area Networks (LAN)

2. Metropolitan Area Network (MAN)
3. Wide Area Network (WAN)

a) **Local Area Network (LAN)** is a privately owned network that serves users within a single building as an office, or a group of buildings close together as a college campus.

Characteristics include:

- ✓ Limited geographical area less than two kilometers.
- ✓ High bandwidth greater than 1mbps.
- ✓ Inexpensive cable media (coaxial or twisted pairs).
- ✓ Good for resource sharing.

b) **Metropolitan Area Network (MAN)** is a communication network covering a large geographic area when compared to LAN, like a city or suburb. A MAN is basically bigger version of a LAN covering a group of nearby corporate offices in a city and might be either private or public. Multi Campuses University can link LAN in the various campuses together to form MAN.

c) **Wide Area Network (WAN)** is communication Network spanning a huge geographical area like a state, country or a continent. It contains a collection of machines intended for running user (i.e., application) programs. Characteristics include:

- ♣ No geographical limit
- ♣ Can connect computers and other devices in different parts of the world
 - **Examples:**
 - ✓ A corporation with offices in London & New York
 - ✓ The Internet

Note: Whatever the type of the Network, they have the following components and features in common:

- ✓ **Servers:** Computers that provide shared resources to the Network users.
- ✓ **Clients:** Computers that access shared resources provided by servers.
- ✓ **Media:** The way in which computers are connected.
- ✓ **Resources:** files, printers or other items to be used by network.

4.3. Advantage and Disadvantage of Computer Network

Advantages of Networks

a) **Sharing of peripheral devices:** Laser Printers, Hard disk drives and Scanners are examples of peripheral devices. All these devices cannot be connected to each and every computer as they are very expensive. Hence, these devices can be shared by many users through a network.

b) **Sharing of Programs and data:** In many organizations, people use almost the same software and access the same data. If these software and data were made available for individual users

separately, then it is wastage of memory. Hence, by sharing the software and data the usage of memory can be minimized, thereby minimizing the expense.

c) Better Communication: In the digital world, information can be transferred within fraction of a second over long distances. And one of the greatest features of networks is Electronic Mail (E-Mail) system. Thus, a company can eliminate the delays encountered with standard postal services or with telephone tag, by using E-Mail services.

d) Security of information: Before networks became a common place, an individual employee was storing information in his or her desktop computer which was not secured. But today, such data or information could be backed-up or duplicated on a network storage device, shared by others.

Disadvantages of computer networks

- ♣ Lack of robustness.
- ♣ Spread of computer virus.
- ♣ Independence issues.
- ♣ Health issues.

1. Lack of robustness

Computer networks rely on the main server called the central server. If the central server malfunctions or there is an issue in the central server, then the entire network would stop functioning. So, this is a major disadvantage due to dependency on a single server.

2. Spread of computer virus

As computers in a network are interconnected, there is a high probability that if one of the computers is affected by the virus, others too can get affected. This spread can actually damage the entire system. Also, if the central server gets corrupted, then it's quite dangerous as the network depends on the central server.

3. Independence issues

The entire networking system is based on a centralized server, so it lacks independence. So, individual users lack the freedom to use the computer as per their wish.

4. Health issues

Computer network provides access to a gamut of services including entertainment, gaming, and movies. These result in making you addicted to the content and thus results in overuse of these services. This excessive screen time makes you feel lethargic, causes eye strain and body pain.

Applications of computer network

Computer Network has many advantages in a myriad of domains. We have provided you with some of the popular uses across domains:

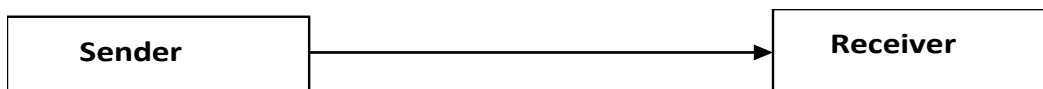
E-commerce	Buying and selling items online
Communication	Video calls
	e-mails
	File transfer
Gaming	Quiz games
Education	Information on the World Wide Web (www) helps students

4.4. Modes of Data Communication

Data communication happens in the form of signals between two or more computing devices or nodes. Data communication between different devices are broadly categorized into 3 types: Simplex communication, Half duplex communication, and Full-duplex communication.

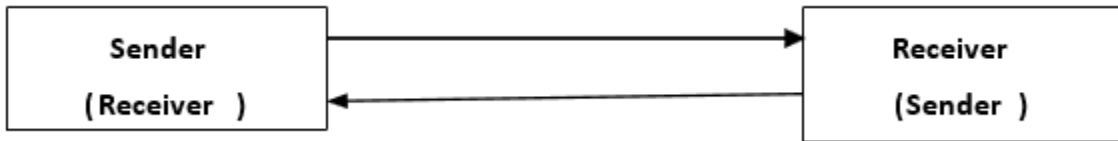
Simplex: In this mode, transmission can take place only in one direction. The device at either end is either a *send-only* or *receive-only* device. E.g., Radio, TV

The flow of data is unidirectional.



Half-duplex: In this mode, data can be transmitted in both directions, but in only one direction at a time. This is good for transmissions in which case the receiver has to acknowledge that it has received the message back to the sender. E.g., Talk back Radio (Walkie talkie)

Thus, a half-duplex line can alternately send and receive data. Two wires are usually required.



Full-duplex: Full-duplex transmission mode allows simultaneous transmission in both directions. When high speed transmission is required, full-duplex transmission is the ideal choice. This transmission mode requires four wires. E.g., Mobile



The primary difference between the three modes of transmission is that in

- ✚ **A simplex mode** of transmission the communication is unidirectional, or one-way; whereas in
- ✚ **The half-duplex** mode of transmission the communication is two-directional, but the channel is interchangeably used by both of the connected devices. On the other hand, in
- ✚ **The full-duplex** mode of transmission, the communication is bi-directional or two-way, and the channel is used by both of the connected devices simultaneously.

Comparison Chart

Basis for Comparison	Simplex	Half Duplex	Full Duplex
Direction of Communication	Unidirectional	Two-directional, one at a time	Two-directional, simultaneously
Send / Receive	The sender can only send data	The sender can send and receive data, but one at a time	The sender can send and receive data simultaneously
Performance	Worst performing mode of transmission	Better than Simplex	Best performing mode of transmission
Example	Keyboard and monitor	Walkie-talkie	Telephone

4.5. Transmission Media

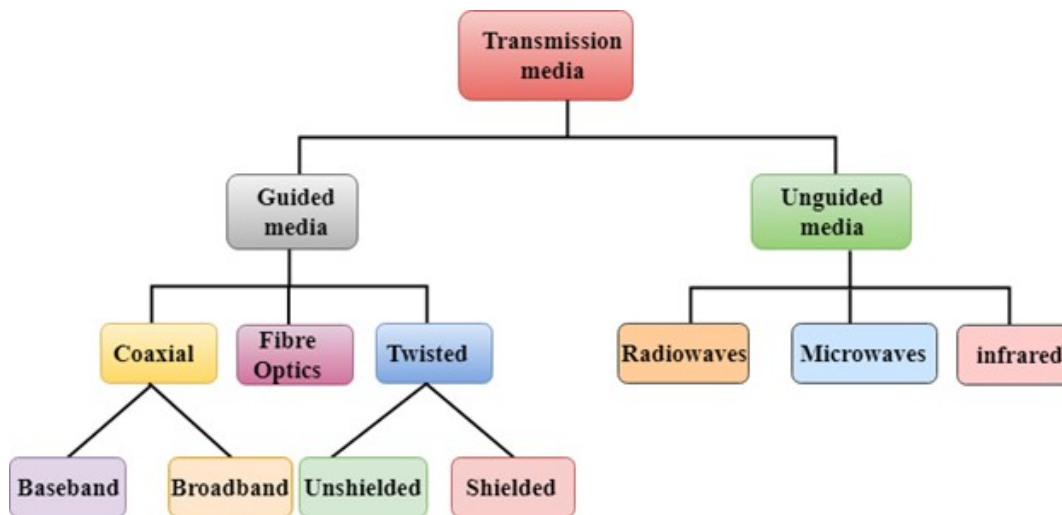
- ♣ Whenever we talk about communication between two computing devices using a network, **five** most important aspects come to our mind. These are
- ♣ Sender, receiver, communication medium, the message to be communicated, and certain rules called protocols to be followed during communication.
- ♣ The communication media is also called transmission media. Components of data communication.
- ♣ **Sender:** A sender is a computer or any such device which is capable of sending data over a network. It can be a computer, mobile phone, smartwatch, video recording device, etc.
- ♣ **Receiver:** A receiver is a computer or any such device which is capable of receiving data from the network. It can be any computer, printer, laptop, mobile phone, television, etc
- ♣ **Message:** It is the data or information that needs to be exchanged between the sender and the receiver. Messages can be in the form of text, number, image, audio, video, multimedia, etc.
- ♣ **Communication (transmission) media:** It is the path through which the message travels between source and destination. It is also called medium or link which is either wired or wireless. For example, a television cable, telephone cable, ethernet cable, satellite link, microwaves, etc.
- ♣ **Protocols:** It is a set of rules that need to be followed by the communicating parties in order to have successful and reliable data communication. You have already come across protocols such as Ethernet and HTTP.

4.5.1. Transmission media

- ♣ A transmission medium can be anything that can carry signals or data between the source (transmitter) and destination (receiver).
- ♣ In data communication, transmission media are the links that carry messages between two or more communicating devices.
- ♣ It is a physical path for data transfer through electromagnetic signals.
- ♣ It is a physical path between transmitter and receiver in data communication.

- ♣ Transmission media is of two types are wired (guided) media and wireless (unguided) media.

Classification of Transmission Media:



- a) **In guided transmission**, there is a physical link made of wire/cable through which data in terms of signals are propagated between the nodes. These are usually metallic cable, fiber-optic cable, etc. They are also known as wired media.

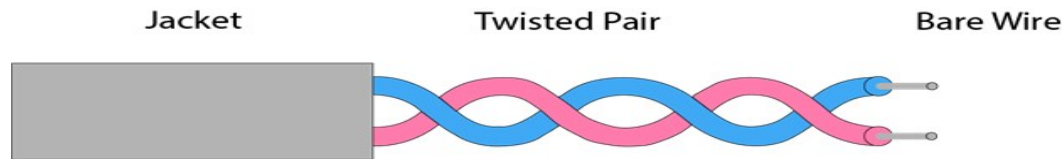
♣ Features:

- ✓ High Speed
- ✓ Secure
- ✓ Used for comparatively shorter distances

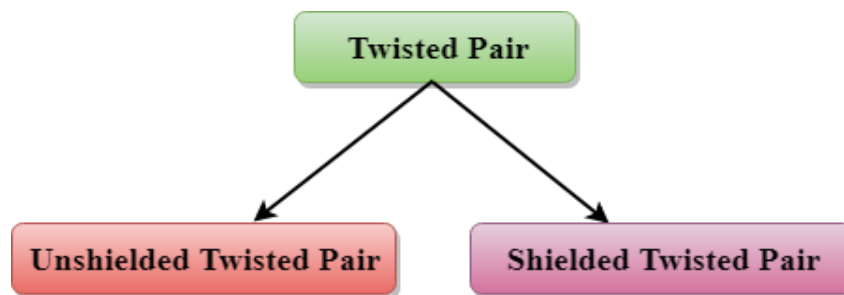
There are 3 major types of Guided Media:

Twisted pair:

Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable. The frequency range for twisted pair cable is from 0 to 3.5KHz.



Types of Twisted pair:



Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication.

Advantages of Unshielded Twisted Pair:

- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

Shielded Twisted Pair

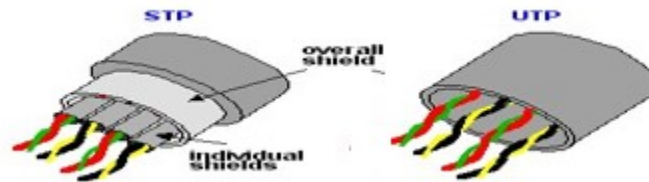
A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Characteristics Of Shielded Twisted Pair:

- ♣ The cost of the shielded twisted pair cable is not very high and not very low.
- ♣ An installation of STP is easy.
- ♣ It has higher capacity as compared to unshielded twisted pair cable.
- ♣ It is shielded that provides the higher data transmission rate.

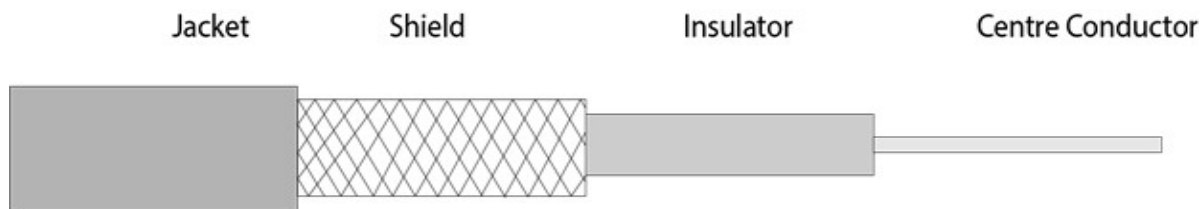
Disadvantages

- ♣ It is more expensive as compared to UTP and coaxial cable.
- ♣ It has a higher attenuation rate.



Coaxial Cable

- ♣ Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- ♣ The name of the cable is coaxial as it contains two conductors parallel to each other.
- ♣ It has a higher frequency as compared to twisted pair cable.
- ♣ The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- ♣ The middle core is responsible for the data transferring whereas the copper mesh prevents from the **EMI** (Electromagnetic interference).



Coaxial cable is of two types:

1. **Baseband transmission:** It is defined as the process of transmitting a single signal at high speed.
2. **Broadband transmission:** It is defined as the process of transmitting multiple signals simultaneously.

Advantages of Coaxial cable:

- ♣ The data can be transmitted at high speed.
- ♣ It has better shielding as compared to twisted pair cable.
- ♣ Highly insensitive to EMI
- ♣ It provides higher bandwidth.

Disadvantages of Coaxial cable:

- ♣ It is more expensive as compared to twisted pair cable.

- ♣ Although fairly insensitive to EMI, coax remains vulnerable to EMI in harsh conditions such as factories.

Fiber Optic

- ♣ Fiber optic cables utilize light waves to transmit data through a thin glass or plastic fiber.
- ♣ Fiber optic cables use light pulses instead of electrical signals, they offer very high bandwidth
- ♣ Because the signal in a fiber optic cable consists of light pulses, the signal cannot be affected by electromagnetic interference. Nor can the cables radiate radio frequency noise.
- ♣ Optical fibers are, therefore, suitable for use in the noisiest and most sensitive environments.
- ♣ Fiber optic transmissions are extremely secure.
- ♣ Installation of fiber optic cable requires greater skill than is necessary to install most copper cables.
- ♣ Fiber optics provide faster data transmission than copper wires.

Summary of Cable Characteristics				
Cable Type	Cost	Installation	EMI Sensitivity	Data Rate
UTP	Low	Low	High	Low
STP	Medium	Medium	Low	Medium
Coax	Medium	Medium	Low	High
Fiber Optic	High	High	None	Very High

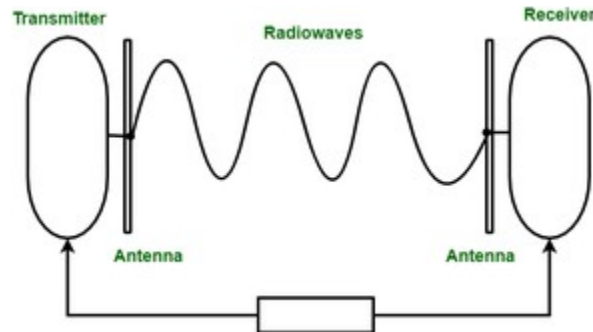
- b) In unguided transmission, data travels in air in terms of electromagnetic waves using an antenna. They are also known as wireless media. No physical medium is required for the transmission of electromagnetic signals.

Features:

- The signal is broadcasted through air
- Less Secure
- Used for larger distances

Examples of Signals transmitted through unguided media:

(I) **Radio waves** – These are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range: 3KHz – 1GHz. AM and FM radios and cordless phones use Radio waves for transmission.



(II) **Microwaves** – It is a line-of-sight transmission i.e., the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz – 300GHz. These are majorly used for mobile phone communication and television distribution.

(III) **Infrared** – Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems. Frequency Range: 300GHz – 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.

4.6. Network Topology

The way in which connections are made is called the **topology** of the network.

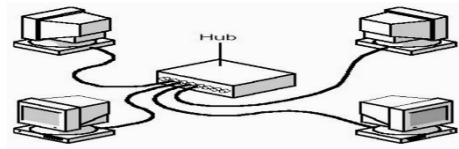
Network topology specifically refers to the physical layout of the network, especially the locations of the computers and how the cable is run between them.

Four of the most common topologies are bus, star, ring, and mesh.

The following five types of physical topologies are frequently used in computer networking:

1. Star Topology
2. Bus Topology
3. Ring Topology
4. Mesh topology
5. Hybrid Topology

1) Start Topology



- ☐ In a star topology all stations are wired to a central wiring concentrator called a hub.
- ☐ Similar to a bus topology, packets sent from one station to another are repeated to all ports on the hub.
- ☐ This allows all stations to see each packet sent on the network, but only the station a packet is addressed to pays attention to it.
- ☐ In a star topology, each station is connected to a central wiring concentrator, or hub, by an individual length of twisted pair cable.
- ☐ The cable is connected to the station's NIC at one end and to a port on the hub at the other.

Advantages of Stars topology

Most modern cabling systems are designed in a star physical topology. The benefits of the star topology are:

- ♣ Each device is isolated on its own cable.
- ♣ It is a reliable network and easy to modify
- ♣ Easy to add new workstation.
- ♣ A node can fail without affecting other nodes
- ♣ All data goes through the central point, which can be equipped with diagnostic devices that make it easy to trouble shoot and manage the network.

Disadvantages of Star topology

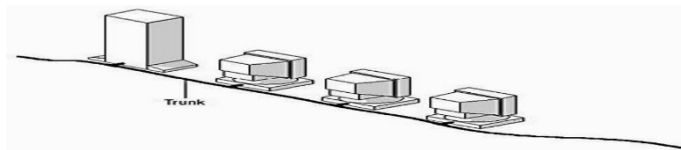
Star topology has the following disadvantages:

- ♣ Because point-to-point wiring is utilized for each node, more cable is required.
- ♣ Hub failures results in a short down of the entire network.

2) Bus Topology

- ☐ In a bus topology, all devices attach to the same transmission medium.
- ☐ The medium has a physical beginning and end.

- ❑ All buses are implemented using electrical cable, usually coax, and the ends of the cable must be terminated with a *terminating resistor* that matches the *impedance* of the cable.
- ❑ The terminating resistor prevents data reflections from coming across as data corruption.
- ❑ The bus is considered a multipoint system because all devices tap into the same *backbone cable*.
- ❑ An important characteristic to remember in bus topologies is that all data signals are broadcast throughout the bus structure.
- ❑ If node B sends a signal to node C, the signal propagates for the length of the cable and is seen by nodes A and D as well.



Advantages of Bus topology

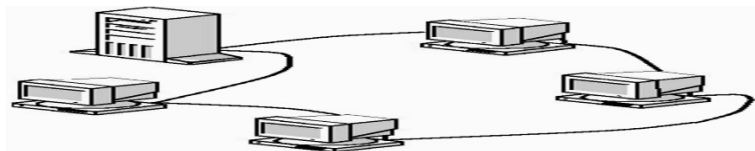
- Bus topology has the following advantage:
 - ♣ Cabling costs are minimized because of the *common trunk*.

Disadvantages of Bus topology

- ♣ Difficult to trouble shoot because no central distribution points exist.
- ♣ Cable breaks can disable the entire segment because they remove the required termination from each of the two cable fragments.

3) Ring Topology

- ❑ The ring topology is a physical, *closed loop* consisting of point-to-point links.
- ❑ Each node on the ring acts as a *repeater*.
- ❑ It receives a transmission from the previous node and amplifies it before passing it on.



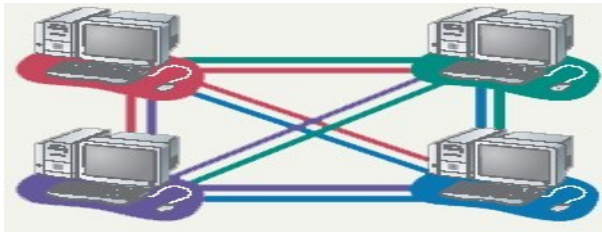
Benefits of Ring topology

- Ring topology has the following advantage:
 - ♣ Each repeater duplicates the data signals so very little signal degradation occurs.

Disadvantage of Ring Topology

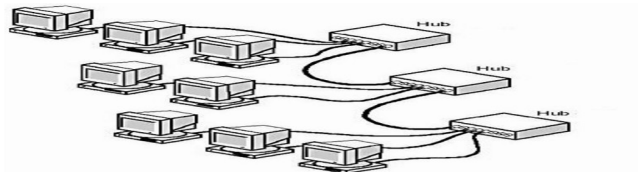
- ♣ A break in the ring can disable the entire network.
- ♣ Because each node must have the capability of functioning as a repeater, the networking devices tend to be more expensive.

4) Mesh Topology



- All computers connected together
- Internet is a mesh network
- ♣ **Advantage**
 - Data will always be delivered
- ♣ **Disadvantages**
 - Lots of cable
 - Hard to setup

5) Hybrid Topology



- The *hybrid topology* scheme combines multiple topologies into one large topology.
- The hybrid network is common in large wide-area networks. Because each topology has its own strengths and weaknesses, several different types can be combined for maximum effectiveness.

Advantages of Hybrid topology

- One company can combine the benefits of several different types of topologies.
- Workgroup efficiency and traffic can be customized.

Disadvantages of Hybrid topology

- Devices on one topology cannot be placed into another topology without some hardware changes.

4.7. Introduction to the Internet

Internet is a global communication system that links together thousands of individual networks. It allows exchange of information between two or more computers on a network. Thus internet helps in transfer of messages through mail, chat, video & audio conference, etc. It has become mandatory for day-to-day activities: bills payment, online shopping and surfing, tutoring, working, communicating with peers, etc. Internet was evolved in 1969, under the project called ARPANET (Advanced Research Projects Agency Network) to connect computers at different universities and U.S. defense. Soon after the people from different backgrounds such as engineers, scientists, students and researchers started using the network for exchanging information and messages.

4.8. Services of the Internet

4.8.1. E-mail

Electronic mail, or ‘email,’ is a digital communication method that uses electronic devices to deliver messages. Electronic mail, commonly shortened to “email,” is a communication method that uses electronic devices to deliver messages across computer networks. "Email" refers to both the delivery system and individual messages that are sent and received.

4.8.2. World Wide Web

The World Wide Web (WWW), commonly known as the Web, is an information system enabling documents and other web resources to be accessed over the Internet. World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc. Users can access the content of these sites from any part of the world over the internet using their devices such as computers, laptops, cell phones, etc. The WWW, along with internet, enables the retrieval and display of text and media to your device.

4.8.3. The remote login/ execution

Remote access, also known as remote login, is the ability to access the data stored on a computer from a remote location. It enables you to open, edit, and save files located on your device from

anywhere in the world. This ability is handy for offsite workers, travelers, and those who work out of office.

4.8.4. The video conferencing

Video conferencing is an online technology that allows users in different locations to hold face-to-face meetings without having to move to a single location together. This technology is particularly convenient for business users in different cities or even different countries because it saves time, expenses, and hassles associated with business travel. Uses for video conferencing include holding routine meetings, negotiating business deals, and interviewing job candidates. Video conferencing is live, visual connection between two or more remote parties over the internet

4.8.5. Search Engine

It is an information retrieval system designed to help find information stored on a computer system, such as on the World Wide Web, inside a corporate or proprietary network, or in a personal computer. The search engine allows one to ask for content meeting specific criteria and retrieves a list of items that match those criteria.

Advantages of Internet

- **Getting Information Right Away.** The internet is super-fast when it comes to finding information
- **Learning Resources.** The internet is a great place to learn new things whenever you want.
- **Making Life Easier.**
- **Easy Communication.**
- **Socializing.**
- **Source of Entertainment.**
- **Source of Earning.**
- **Staying Informed.**

Dis Advantages of Internet

Reading assignment